

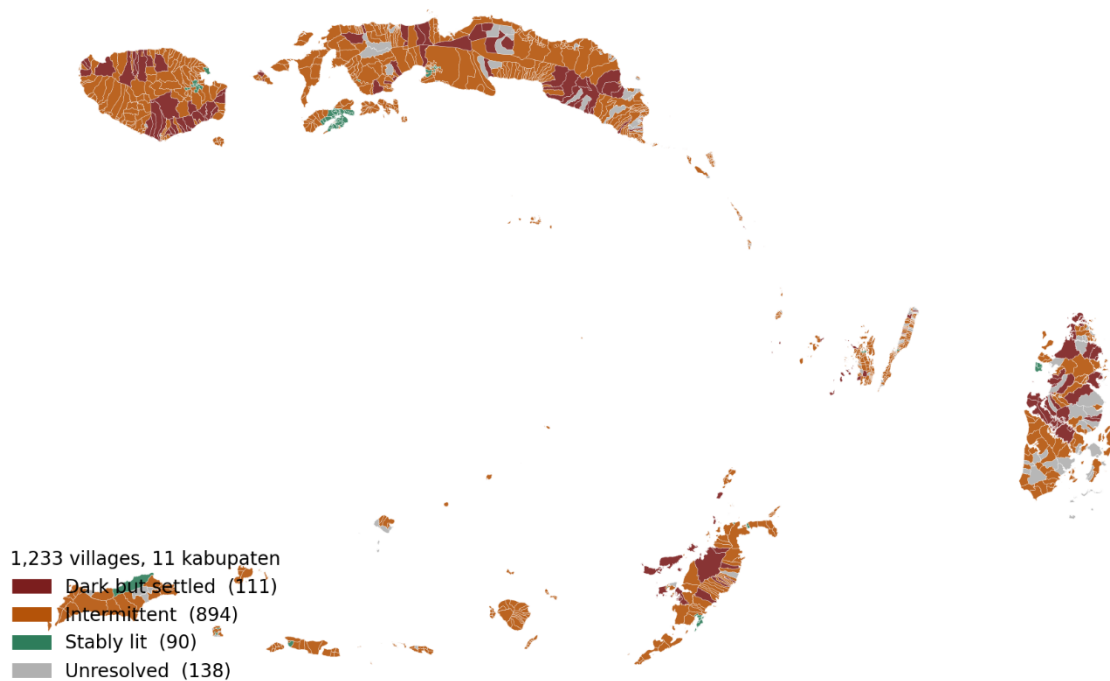
Hackathon Proposal

Executive Summary

While Indonesia reports an official village electrification ratio of 99.90%, national utility figures reveal that only 92.75% of villages are connected to the grid. The remaining gap is filled by unmonitored solar lanterns or small diesel systems that quickly fail, leaving over 4,808 unconnected villages—concentrated primarily across eastern Indonesia and archipelagic ASEAN—invisible to planners, utilities, and investors. While macro-initiatives like ASEAN Power Grid (APG) focus on interconnecting national grids across mainlands, Terang addresses the millions of people in island communities that exist beyond the physical and economic reach of any grid.

We have tested this approach in Maluku province, analyzing all 1,233 villages using free satellite nighttime-light imagery. By replacing standard built-up pixel masking with an unmasked polygon-based reduction—recovering the 79% of remote settlements typically lost in standard geospatial analyses—we identified 111 settled villages emitting light at or below uninhabited baseline levels, alongside 894 villages suffering from severe, seasonal service intermittency (Figure 1).

Maluku village electrification classes — Layer 1 output



Source: Terang Layer 1 pipeline v3 · VIIRS 2024-07 to 2026-04 · BIG boundaries

Figure 1. Maluku province gap map. All 1,233 villages coloured by detection class – dark but settled (red), intermittent (amber), stably lit (green), unresolved (grey outline).

Terang translates these satellite observations into an end-to-end, investable off-grid pipeline through a three-layer architecture:

- **Layer 1 (Detect):** Uses VIIRS nighttime-light data and settlement footprints to pinpoint dark and intermittent villages, offering cheap, continuous, independent observation where official records fail.
- **Layer 2 (Prescribe):** Applies an auditable, rule-based decision tree that screens sites for environmental and policy constraints (directing protected-area overlaps into policy pathways and grid-proximate sites to utility extension) before matching remaining dark villages to the least-cost technology indicated by local geography (e.g., run-of-river micro-hydro, solar-biomass, or solar-battery systems).
- **Layer 3 (Pool & Verify):** Bundles individually unbankable village mini-grids into district-level portfolios for institutional blended finance. It replaces costly field audits with continuous satellite verification,

unlocking milestone disbursements—including a performance tranche for productive-use signatures—only when light is confirmed on and sustained. Crucially, it translates technical and routing feasibility into bankable investment opportunities by pairing financial metrics with Social Cost Benefit Analysis (SCBA) and Environmental Benefit Analyses quantifying energy savings, income generation, and carbon offsets to de-risk capital deployment beyond the grid.

Beyond technical verification, Terang operates as a sustainable social enterprise. By leveraging open, global datasets—requiring only country-specific boundary files—the framework is immediately scalable to comparable island geographies across ASEAN, with the Philippines as a natural second deployment target. By charging verification fees priced significantly lower than traditional field audits, Terang channels a ring-fenced portion of its revenue into a community-managed Village Reliability Fund to pre-finance repairs, while hiring paid local verifiers to validate satellite findings. Ultimately, Terang bridges the gap between capital and off-grid communities, ensuring that energy investments yield verifiable, long-term socioeconomic and environmental resilience.

Introduction

Background

The Directorate General of Electricity reports a villages electrification (in Indonesia) ratio of 99.90% (Figure 2), while the same figures show only 92.75% of villages served by PLN, the national utility. The gap between those numbers is where this proposal begins. (ESDM, 2023) A village counted as electrified may have a 24-hour grid connection, or a batch of solar lanterns distributed five years ago. Both are recorded identically, and when lantern or diesel systems fail within two to three years, no institutional record captures the reversal.

The shortfall is concentrated in eastern Indonesia: 4,808 villages have no PLN connection, 4,398 of them in Papua and the rest across Maluku, Nusa Tenggara and Sulawesi, the small, dispersed, maritime communities, expensive to survey and easy to overlook. (ESDM-Electrification Statistics, 2023; PLN Annual Report, 2023) Because the statistics say the problem is nearly solved, no data trail identifies which villages remain unserved. Without that list, budgets are allocated on aggregates and investors cannot readily identify projects. We have selected Maluku as the pilot province rather

than Papua because its boundary data is complete, its scale permits an end-to-end run within the time available, and its archipelagic conditions are broadly representative of the larger gap in Papua and of comparable island geographies elsewhere in ASEAN. Papua is the intended target for national replication rather than for initial validation.

The claim gap: reported electrification vs. actual grid connection

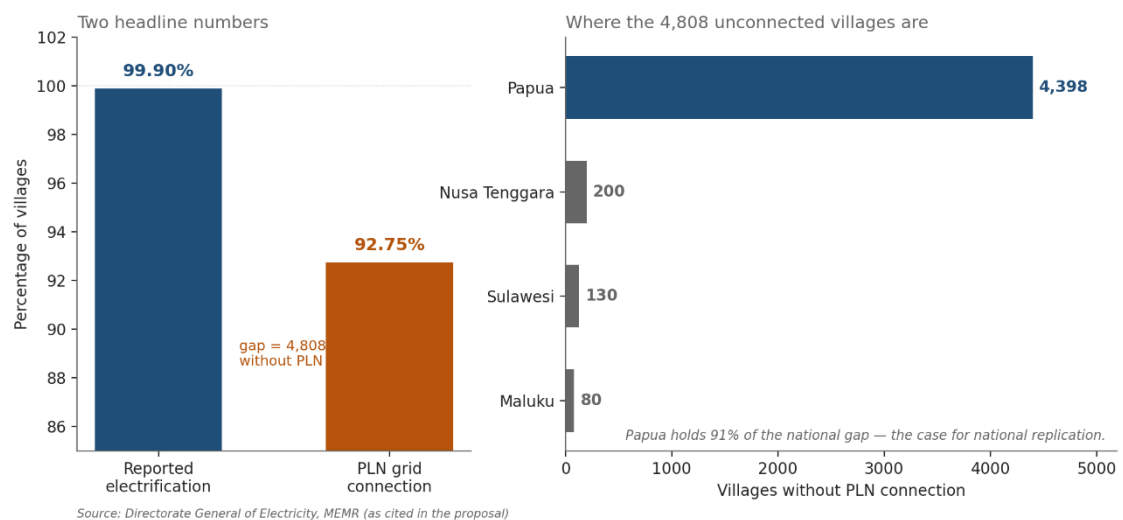


Figure 2. Reported village electrification ratio (99.90%) against villages served by PLN (92.75%), with the 4,808 unconnected villages disaggregated by province.

Relevance

ASEAN is preparing to invest in cross-border transmission through the ASEAN Power Grid and recent analyses highlight that this project will need approximately \$100 billion USD (Ember Report, 2025; Eco-Business, 2025). Modernizing these corridors is widely recognized as the primary catalyst to unlock clean energy investment.

However, ASEAN's energy challenge extends far beyond connecting the mainlands—the region is home to vast archipelagic nations like Indonesia and the Philippines, where the main reason remote communities lack electricity is not neglect, but brutal terrain, prohibitive grid extension costs, or unaffordable standard tariffs.

This is where Terang address energy security and accessibility beyond the grid. Highlighted at SAYEF 2026, Terang locates communities that current systems cannot see—pinpointing named villages with exact coordinates and custom recommendations. On energy security, it advocates for distributed generation in remote islands to replace price-volatile, storm-vulnerable

diesel shipments with resilient local power, utilizing satellite monitoring for real-time early warning when outages occur.

While the APG program connects national grids to one another, Terang addresses the communities beyond every grid. Because its core methodology relies on free, global datasets, requiring only country-specific village boundary files, it is readily scalable across the region, making island-heavy nations like the Philippines natural candidates for deployment.

Objectives

- 1. Detect.** Identify villages where official claims and satellite observation disagree, at village level rather than in aggregate, using only free data.
- 2. Prescribe.** Match each flagged village to the least-cost technology its geography appears to support, after separating sites that fall within legally protected areas.
- 3. Pool.** Aggregate individually unbankable village projects into district portfolios large enough for institutional capital.
- 4. Verify.** Replace slow, costly field verification with continuous satellite monitoring, so milestones can be confirmed cheaply and repeatedly.
- 5. Sustain.** Introduce sustained-service financing: payment for light that persists, not for connections installed.
- 6. Quantify.** Express project value not only in financial returns but in a structured socioeconomic and environmental benefit framework, covering employment, household welfare, emission reduction, and electrification reliability, so that concessional capital can be justified on development grounds alongside commercial ones.
- 7. Share.** Publish the methodology openly, including its limitations, for reuse across ASEAN.

Problem Statement

Rural electrification in archipelagic Southeast Asia fails at three points, and each failure compounds the next.

Measurement. Statistics record connections, not service. A village with three broken lanterns and a village with 24-hour grid power appear identical. Because that choice determines where budgets go, the communities most in need are hidden by the figures meant to describe them.

Origination. A single village mini-grid may cost 200,000 US dollars (ESMAP, 2019), far below the minimum ticket for institutional investors. Facilities mandated to fund energy access consistently report difficulty finding verified,

bankable projects. Capital and need coexist with no mechanism to connect them.

Persistence. Finance pays on installation. Once a system is commissioned nobody is paid to confirm it still works, and field verification consumes 10 to 15 per cent of programme costs, so it is done once and rarely repeated. The result is systems built, counted, then quietly failing. Terang addresses all three with one capability: cheap, continuous, independent observation.

Proposed Solution

Solution Concept and Description

In plain terms: we use satellites to see which villages are actually dark at night, work out what should be built in each, bundle them into investment packages, and release money only when the lights are confirmed on and staying on. Three layers, each feeding the next as shown in Figure 3.

Layer 1 – Detect. Satellites operated by NASA and NOAA record light emitted at night. We compare this against settlement footprints for every village. Villages that are demonstrably inhabited yet emit light at or below the radiance of uninhabited terrain are flagged for attention. Cross-referencing against official electrification records – which would allow flagged villages to be described as claimed-electrified rather than simply dark – is the next data step and depends on a village-code crosswalk currently under development. Output: a ranked, confidence-scored map of the apparent access gap.

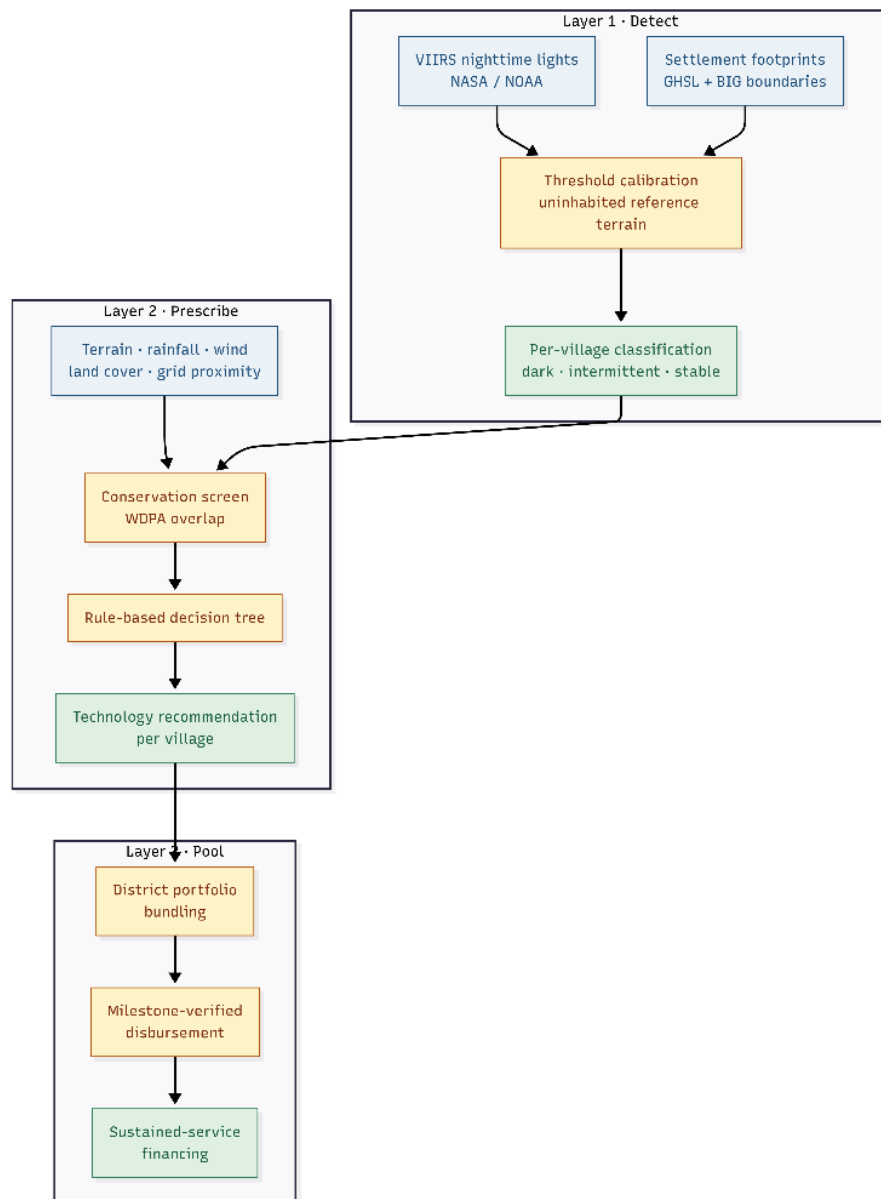


Figure 3. Terang three-layer architecture: Detect → Prescribe → Pool, showing the data inputs to each layer and the outputs passed downstream.

Layer 2 – Prescribe. A transparent decision tree recommends the least-cost technology indicated by terrain, river catchment, rainfall, wind, land cover and distance to the nearest grid-served settlement. The tree evaluates two contextual conditions before any commercial-viability logic. Sites where more than ten per cent of the village area falls within a legally protected area are separated from the commercial pipeline and directed toward a policy pathway: rather than proposing infrastructure inside a national park, these villages are flagged for coordinated engagement between the Ministry of Environment and Forestry, the relevant park authority and the local community, in a process that may take years and that Terang does

not itself undertake. Sites within approximately five kilometres of a reliably lit settlement (ESMAP, 2020; Mentis et al., 2017) are separated from the off-grid pipeline and directed to grid extension, which under Indonesian regulation is PLN's responsibility rather than a private-developer opportunity; Terang's role for these villages is to hand the identified list to PLN and to the regional government, not to design their electrification. Only after these two contextual filters does the tree evaluate hydro potential (IEA-Hydropower Special Market Report, 2021; ESMAP, 2018), wind speed (IRENA-Future of Wind, 2019; Global Wind Atlas 3.0-DTU/World Bank, 2021), biomass feedstock and the solar-battery default. To refine this resource screening, we integrate localized geospatial data from Indonesia Ministry of Energy and Mineral Resources (ESDM)'s Geoportal OneMap-mapping (ESDM One Map, 2024), both regional renewable energy potential and existing onshore conventional energy infrastructure. Incorporating conventional assets allows the model to identify opportunities for asset repurposing or hybrid configurations (such as solar-diesel systems), optimizing existing legacy generation while transitioning remote communities toward reliable, low-carbon power. Extending the conservation screen to intact forest that is present but not formally protected is the next methodological step and will require land-cover time-series analysis. Output: a recommendation per village with indicative size, cost band, and the decision rule that produced it.

Layer 3 – Pool and Verify (Figure 4). Villages are bundled into district portfolios sized for institutional investment and structured for blended finance. Milestones are tied to observable events: baseline darkness before contracting, commissioning detected as a step change in light, and persistence verified over the following months and years. A modifier applies where satellite radiance patterns are consistent with productive load – sustained draw during pre-dawn or post-sunset hours, of the kind produced by cold storage, ice-making, drying racks or milling equipment rather than by lighting alone. Villages exhibiting this signature release an additional persistence tranche to their developer, on the reasoning that productive use is what raises household earnings and therefore willingness and ability to pay.

Beyond technical and resource screening—such as evaluating main-grid extension, off-grid feasibility, community-driven models, or government funding alignment, Terang evaluates comprehensive investment possibilities for financiers. By synthesizing economic viability with Social Cost Benefit Analysis (SCBA) and Environmental Benefit Analyses, the platform quantifies expected household energy savings, income uplift, direct job creation, and

avoided carbon emissions, offering impact investors and development banks a fully de-risked, bankable pipeline.

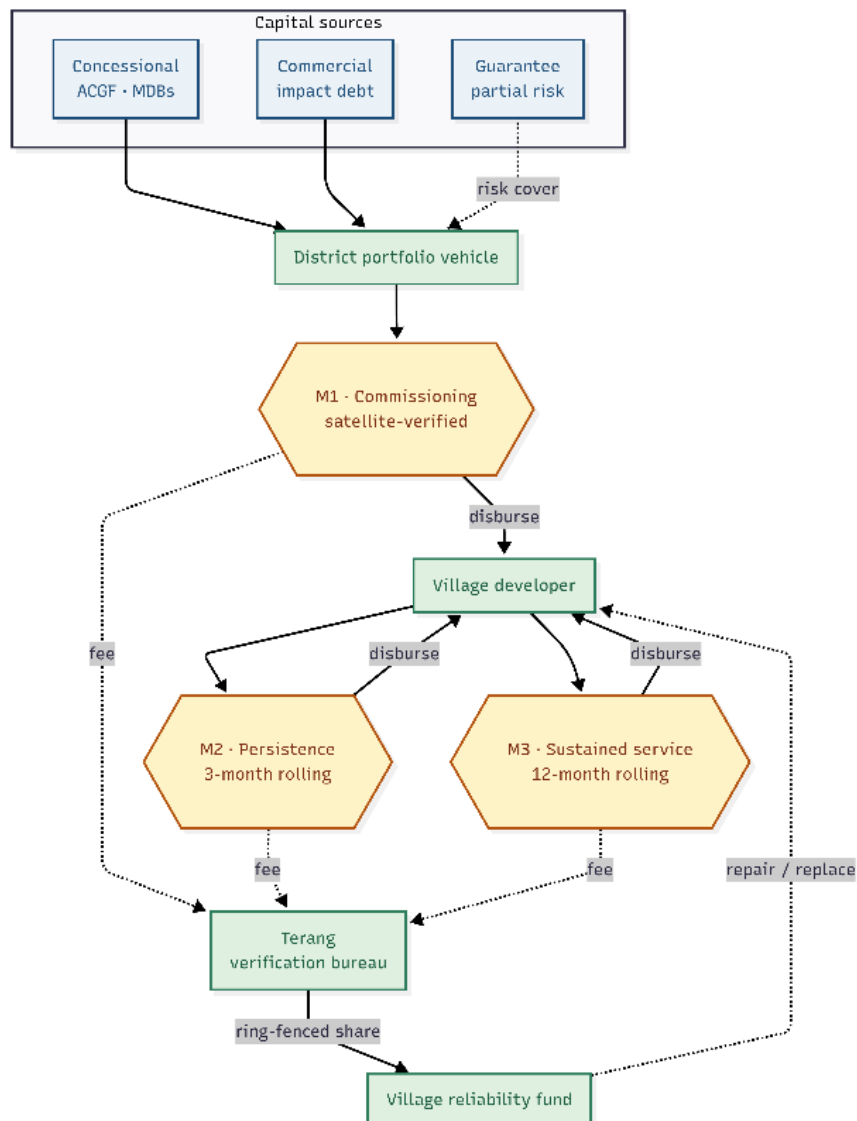


Figure 4. Layer 3 financing structure. Capital sources (concessional, commercial, guarantee) flowing into a district portfolio vehicle; disbursement gated on satellite-verified milestones M1 commissioning, M2 persistence, M3 sustained service; village reliability fund fed by a ring-fenced share of verification revenue.

Scope of Work

Completed. Layer 1 has been built and run on real data for Maluku: 1,233 villages across 11 Regency, 22 months of imagery, with a detection threshold calibrated from uninhabited reference terrain. Layer 2 has been built end-to-end across the 1,005 flagged villages, incorporating resource screening and protected-area screening against the World Database on Protected Areas.

A working demonstration application presents all three layers. Our Layer 1 detection closely aligns with several recent ground reports indicating around 130 unelectrified villages in Maluku (bedahusantara.com, 2025) (matching our 111 dark villages classifications), and ESDM allocation plan for the 2026 Village Electrification program across 1,520 locations (Azura-bloombergtechnoz.com, 2026).

In progress. Screening for intact forest using Hansen land-cover time series, which our present implementation does not yet compute reliably and which we therefore treat as unresolved; population and household counts drawn from WorldPop; the village-code crosswalk that would link radiance measurements to official electrification claims; and deployment to a second province with a larger recorded access gap.

Importantly, our Layer 2 technology prescription pipeline is undergoing active expansion to incorporate a broader array of localized spatial datasets. While baseline resource screening for hydro, wind, biomass, and solar potential is functional, we are currently integrating high-resolution geospatial layers from ESDM's Geoportal OneMap. This includes incorporating regional renewable energy resource mapping and onshore conventional infrastructure data, such as existing diesel assets (PLTD) to evaluate asset repurposing and hybrid configurations like solar-diesel systems. Because these multi-source data streams are still being harmonized, the final decision-tree thresholds and technology mixes remain subject to further calibration.

Finally, the analytical outputs of Layer 2 are being connected to the application's decision-support engine to run full economic and Social Cost Benefit Analysis (SCBA) and Environmental Benefit Analyses. Monetized metrics covering household fuel expenditure savings, productive-use income uplift, local job creation, and carbon offset accounting are currently being coded into the application's backend. Completing these modules will allow the system to automatically generate de-risked impact profiles per village cluster, preparing the platform for full provincial validation in Maluku before expanding to a second pilot province with a larger recorded access gap, such as Papua.

Out of scope. Terang does not build, own or operate generation assets, and does not replace engineering feasibility studies, hydrological survey or advanced environmental impact assessment. Its outputs are screening-level instruments directing where more expensive investigation should go.

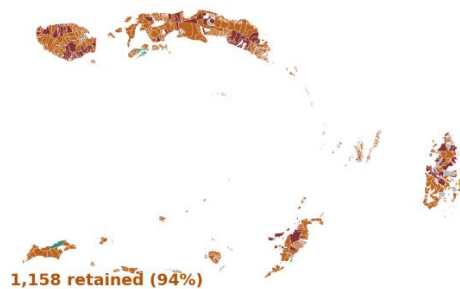
Methodology

Detection. VIIRS is a sensor on polar-orbiting weather satellites measuring light emitted at night. We use 22 monthly composites at roughly 500 metre resolution, with the Global Human Settlement Layer for built-up area and village boundaries from Indonesia's national geospatial agency. The threshold separating lit from unlit is not taken from literature but calibrated per study area, by sampling radiance at points at least two kilometres from any built-up surface and inland from water. For Maluku, this procedure yielded a background floor of $0.201 \text{ nW/cm}^2/\text{sr}$ and a detection threshold of $0.292 \text{ nW/cm}^2/\text{sr}$. Reference points were additionally masked against inland water using the JRC Global Surface Water dataset, so that the threshold reflects genuinely uninhabited terrestrial radiance rather than a mixed land-water noise floor.

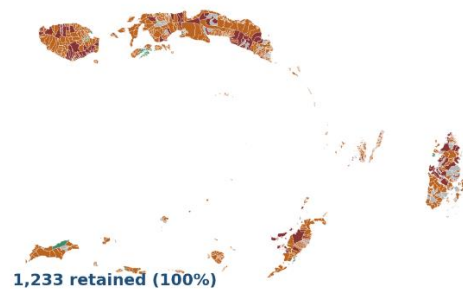
A methodological correction. Standard practice masks radiance to built-up areas before analysis. At VIIRS resolution this eliminated 79 per cent of Maluku's villages, because settlements smaller than one 500 metre pixel lose every pixel to the mask – and those excluded were systematically the smallest and most remote, precisely the population the project exists to find. We corrected it by treating the village polygon as the unit of analysis, keeping settlement data as a size control (Figure 5).

The 79% that standard practice loses

Standard practice: mask VIIRS to built-up at 500 m



Terang correction: reduce over village polygon, unmasked



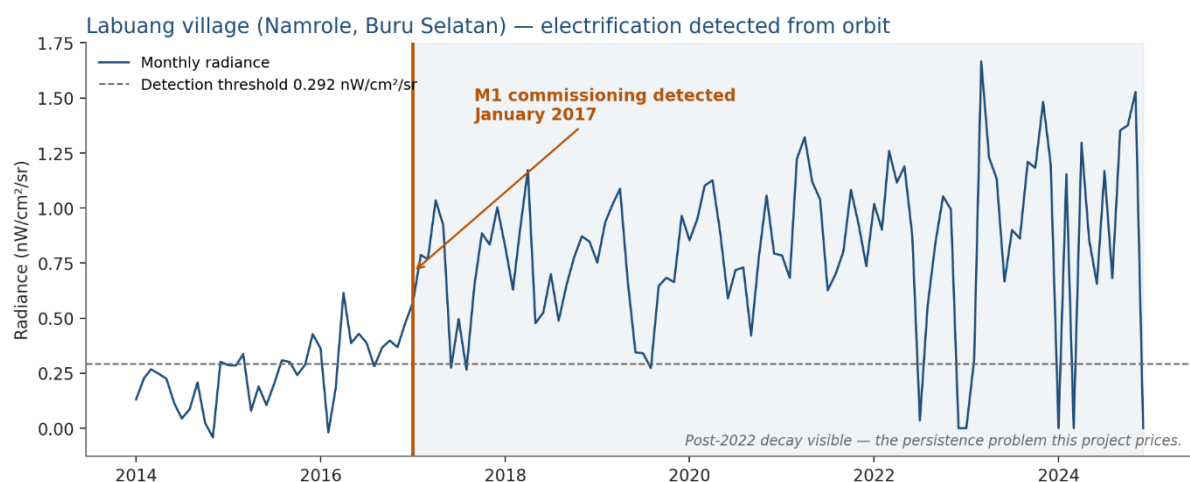
Grey villages on the left are dropped by the mask because no 500 m pixel centroid falls inside them — systematically the smallest and most remote.

Figure 5. Effect of the masking correction. Left: villages retained under standard built-up masking at VIIRS resolution. Right: villages retained under

polygon-based reduction. 79 per cent of villages, systematically the smallest and most remote, are recovered.

Classification and validation. Villages are grouped according to how frequently they emit light above the threshold and how variable that emission is: steady illumination is consistent with grid-quality service, intermittent emission with diesel or degrading solar systems, and absence of emission in a settled area with an access gap. Applied to Maluku, this produced 111 villages classified as dark but settled, 90 as stably lit, and 138 as too small or too frequently cloud-obscured for confident classification, with the remainder classified as intermittent. Within that intermittent group, the proportion of villages emitting light above threshold varies substantially by month and falls sharply during the wet season, a pattern we consider more consistent with genuine service degradation than with measurement noise, though distinguishing the two conclusively would require ground data.

Two checks were applied. The eight brightest villages identified are all located in Ambon city, the provincial capital, at 14 to 29 nW/cm²/sr against a provincial median of 0.251 nW/cm²/sr, which is consistent with expectation. Separately, the commissioning detector – the same function that would trigger disbursements – independently identified January 2017 as the month in which Labuang village in Buru Selatan was electrified, from radiance alone and without ground reporting. That village's light output has visibly degraded since 2022, which we take as indicative of the persistence problem this project seeks to address (Figure 6).



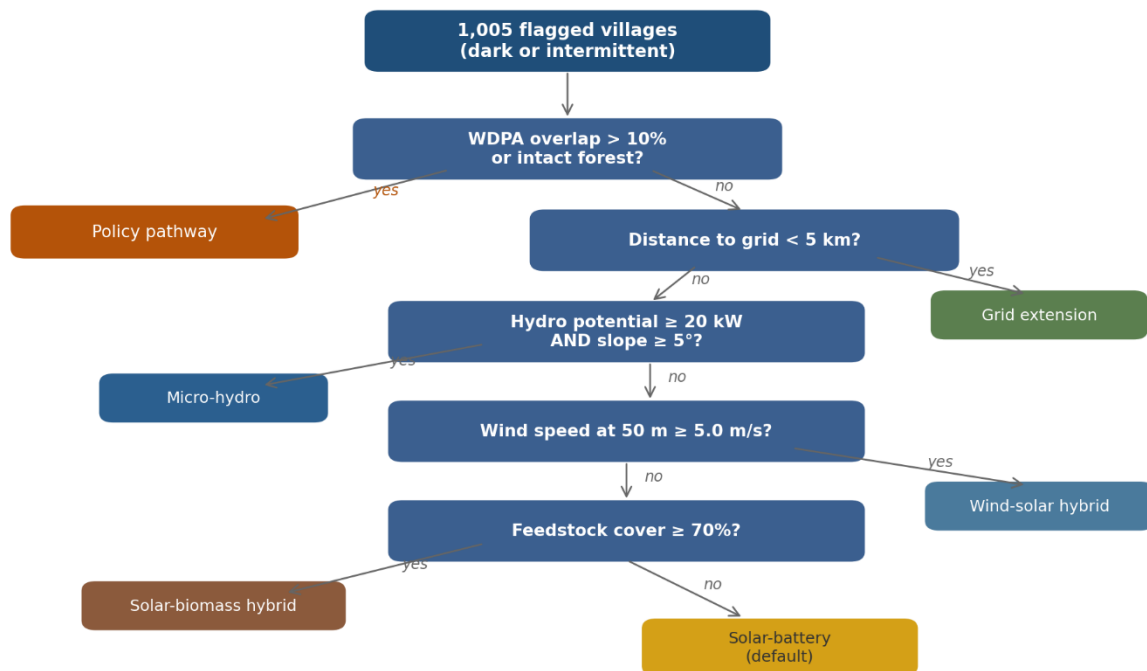
Source: Terang retrospective validation · VIIRS monthly composites 2014-2024

Figure 6. Labuang village, Namrole, Buru Selatan. Monthly radiance 2014-present against the 0.292 nW/cm²/sr detection threshold, with the M1 commissioning step automatically detected at January 2017 and the post-2022 degradation visible.

Related published work, notably the High Resolution Electricity Access project (University of Michigan, with the World Bank and NOAA), offers independent support for the proposition that VIIRS radiance tracks settlement-level electrification (Min, B., et al., 2024). Terang seeks to extend that body of work operationally, translating per-village detection into a technology prescription and a satellite-verified milestone disbursement mechanism.

Prescription. Screening combines elevation and slope, river catchment, rainfall, wind, land cover and distance to the nearest reliably lit settlement. A rule-based decision tree as depicted in Figure 7 assigns technology rather than a statistical model, because a published tree with visible thresholds can be audited and corrected by domain experts. Applied to Maluku's 111 dark villages (Figure 8), 24 were found to fall within protected areas above a ten per cent overlap threshold and were separated from the commercial pipeline for policy engagement. Manusela village, at approximately 50 per cent overlap with Manusela National Park, and three villages in Tanimbar Islands at complete overlap, illustrate the pattern. The remaining 87 villages form the tractable pipeline: 32 micro-hydro candidates, 31 suited to biomass hybrids, 14 within reach of grid extension, 8 solar-battery and 2 wind-solar. We do not assert a ranking within any single technology tier: the highest-scoring micro-hydro sites all saturate the conservative catchment and head caps we apply, and differentiating between them would require field survey.

Layer 2 · technology decision tree



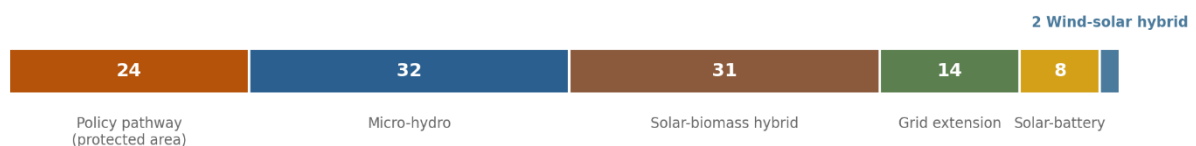
Each branch tests a numeric threshold that is auditable and adjustable by domain experts. The conservation screen fires first, before any commercial-viability logic.

Source: Terang Layer 2 · Cell 19

Figure 7. Layer 2 technology decision tree, showing every branch and threshold in sequence: conservation screen, grid proximity, micro-hydro potential, wind speed, biomass feedstock, solar-battery default.

111 dark villages, mapped to a technology or a pathway

24 villages screened to policy pathway; 87 form the tractable pipeline.



Source: Terang Layer 2 · terang_layer2_prescriptions.csv

Figure 8. Technology prescription mix for Maluku's 111 dark villages: 24 directed to policy pathway (protected area), 14 to grid extension (PLN mandate); the remaining 73 villages form the tractable off-grid pipeline, comprising 32 micro-hydro, 31 solar-biomass hybrid, 8 solar-battery and 2 wind-solar candidates.

Screening also rules options out. Tidal and marine energy, superficially attractive for an island province, remain pre-commercial and unfinanceable at village scale. Solar irradiance across equatorial Indonesia is so uniform it cannot discriminate between sites, so it informs sizing but never site selection. What screening identifies instead is less obvious and more locally

grounded: run-of-river micro-hydro in the high-rainfall uplands of Seram and Buru, and biomass hybrids in the sago and coconut belt.

Limitations. Nighttime lights measure outdoor and aggregate illumination, so an electrified village with very low night-time consumption can appear dark. Radiance confirms village-level energisation and persistence; it does not confirm household connection counts, tariff payment or emissions quantification, each of which needs metered or field data. Terang is a screening instrument that directs ground-truthing, never an audit verdict on any programme or agency.

Pooling. Villages are bundled into district portfolios in two stages: grouping, which decides which villages belong together, and validation, which decides whether the resulting bundle is actually bankable.

Geographic proximity is applied first, using the actual distribution of Maluku's 111 dark villages across its kabupaten: Kepulauan Aru holds 26, the largest concentration, followed by Buru Selatan (19), Seram Bagian Timur (15), and Kepulauan Tanimbar (13) – together, these four kabupaten hold 73 of the 111 villages, well over half the province's total. Kepulauan Aru forms the first bundle because it is the single largest cluster, not an arbitrary selection.

Technology homogeneity follows directly from the Layer 2 decision tree's own thresholds, applied village by village. A bundle is built from villages sharing the same output of this tree – Kepulauan Aru's candidate bundle, for instance, is drawn only from whichever of its 26 villages fall into the same technology branch, not all 26 indiscriminately.

Beneficiary profile is the third grouping criterion. In Kepulauan Aru, the dominant livelihood is small-scale fisheries, which is why its micro-hydro or solar-battery candidates are evaluated for shared cold-storage potential before being finalized into a bundle – the same technology assignment serves a materially different purpose in a fishing-dependent kabupaten than it would in an agricultural one.

Once a candidate bundle is assembled on these three grouping criteria, it is run through the Socioeconomic and Environmental Benefit Analysis before being presented to any financier: quantified household energy savings, income uplift, direct job creation, and avoided carbon emissions. A bundle that clears geography, technology, and beneficiary grouping but scores weakly on economic, SCBA, and environmental benefit is sent back to grouping stage for re-assembly – widening the geographic net to an adjacent kabupaten such as Buru Selatan or Seram Bagian Timur, for example – rather than presented to a financier as-is. Once a bundle clears both stages, capital is drawn into

a single district portfolio vehicle from sources the financiers themselves structure: concessional capital (ACGF, MDBs), commercial impact debt, and a guarantee providing partial risk cover.

Verifying. Disbursement from the vehicle is milestone-gated, and every milestone is a satellite-verified event rather than a field attestation. Commissioning (M1) is detected as a step change in light and releases the first disbursement. From that point, two independent monitoring clocks run concurrently: persistence (M2) is re-confirmed on a rolling three-month basis, and sustained service (M3) on a rolling twelve-month basis – each satisfied window releasing its own disbursement, checked in parallel rather than sequentially. Every M1, M2, and M3 confirmation generates a fee to Terang's verification bureau, distinct from the capital moving to the developer. A satellite flag is a hypothesis, not a fact, until a paid local resident confirms it through a mobile form and an elected village energy committee co-signs the milestone – the only way persistence can be checked monthly rather than once, at a fraction of the cost of a conventional field audit that is typically performed a single time and never repeated. A fixed, ring-fenced share of every verification fee flows automatically into a village reliability fund, which finances repair or replacement directly when ongoing monitoring detects a village has gone dark again – closing the loop between detecting a failure and funding its fix. This same verification layer makes the productive-use modifier possible: a village whose radiance shows sustained pre-dawn or post-sunset draw – cold storage, ice-making, milling – releases an additional persistence tranche once a community verifier confirms the equipment actually exists, since a satellite alone cannot tell a genuine productive-use signature from a false reading of the same data. In Kepulauan Aru specifically, a solar-battery system paired with a shared cold room extends catch shelf life from hours to days, reaching buyers a lighting-only system never could – the same lit-fraction signal from space, a materially different outcome on the ground, and the reason Terang prices the two developer proposals differently rather than treating them the same.

Key Components/Stakeholders

Development financiers. Facilities such as the ASEAN Catalytic Green Finance Facility, the ASEAN Infrastructure Fund and multilateral development banks hold capital mandated for energy access but face a shortage of bankable projects. They are the primary customer, receiving a pre-screened pipeline and a standardised monitoring layer that lowers due-diligence cost per project.

Government and utilities. ESDM, PLN and district governments gain an independent prioritisation instrument for final-mile programmes. The framing matters: Terang helps government reach its own universal access target sooner, rather than auditing past claims.

Mini-grid developers. Qualified leads with indicative technology and sizing, plus faster milestone confirmation, easing the cash-flow constraint verification delays impose today.

Communities and village energy committees. Residents are not passive beneficiaries. Satellite flags are hypotheses; community members confirm or reject them through a simple mobile form and are paid per verification, and elected committees co-sign milestone confirmations. This resolves the method's main technical weakness, creates local ownership, and generates ground-truth data that improves the system over time.

Researchers and civil society. The methodology, including its limitations and the masking correction above, is published openly so it can be reproduced and extended.

Impact and Sustainability

Impact potential. As shown in Figure 9, in Maluku, 111 villages appear settled yet effectively dark, concentrated in Aru Islands (26), Buru Selatan (19), Seram Bagian Timur (15) and Tanimbar Islands (13), with the majority of remaining villages showing unstable service. Nationally, government figures identify 4,808 villages without a PLN connection. The method should transfer to any jurisdiction for which a village boundary file exists.

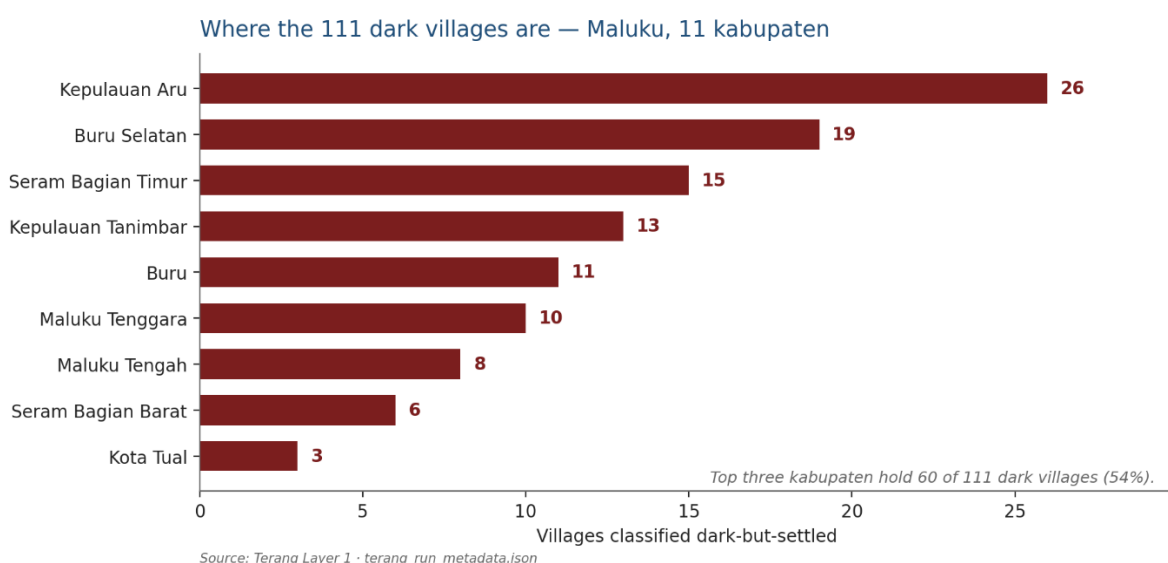


Figure 9. Dark-but-settled villages by Regency: Aru Islands 26, Buru Selatan 19, Seram Bagian Timur 15, Tanimbar Islands 13, Buru 11, Maluku Tenggara 10, Maluku Tengah 8, Seram Bagian Barat 6, Kota Tual 3.

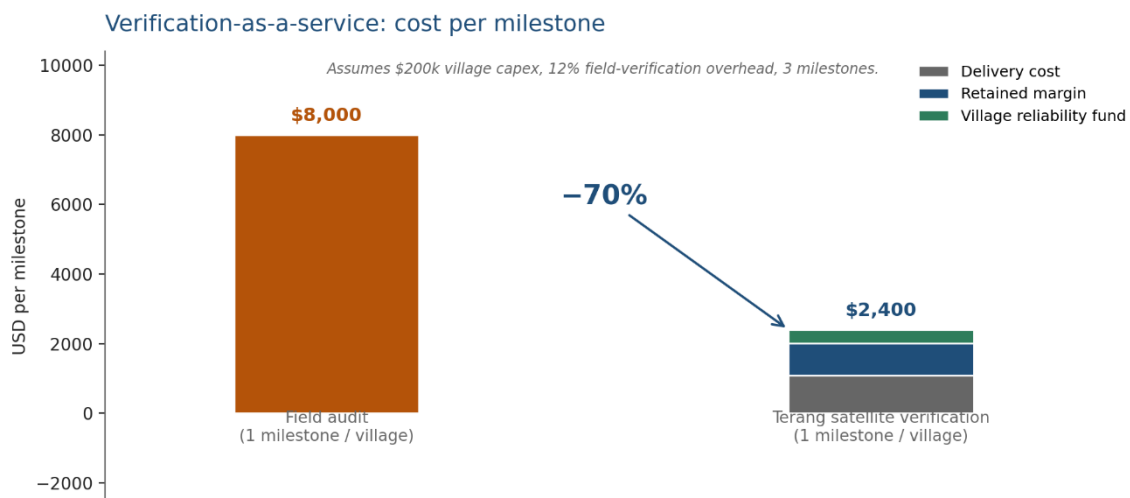
Socioeconomic and environmental impact. Renewable energy projects in remote archipelagic settings are rarely justified by financial returns alone. Terang addresses this by computing an SCBA and environmental benefit for each village cluster (Table 1), translating satellite-observed electrification into monetised and non-monetised impact across three dimensions. Economically, reliable electricity enables productive-use applications – cold storage, fish-drying, milling, ice-making – that raise household incomes and extend market reach beyond what lighting alone achieves; local construction and operation of mini-grid systems also generate direct employment. From a welfare standpoint, replacement of kerosene and diesel systems removes a fuel-cost burden that can represent ten to twenty per cent of household expenditure in off-grid communities (Bacon, Bhattacharya & Kojima, 2010; BPS Susenas 2023), while extended evening hours support education and health outcomes observable over multi-year radiance records. Environmentally, the conservation screen built into Layer 2 diverts projects away from protected and intact forest – each village prescription carries an implied figure for avoided infrastructure encroachment; and where solar-battery or micro-hydro systems displace existing diesel generators, the displaced fuel volume translates to a calculable avoided-emission estimate per village per year. The goal of this project is therefore not simply to increase the number of villages counted as electrified, but to improve the reliability and quality of electrification service – and to demonstrate that doing so produces measurable societal returns that justify the concessional share of blended-finance structures.

SCBA and environmental impact matrix – illustrative per village class

Benefit dimension	Dark settled (111)	Intermittent (894)	Existing PLTD (diesel)	Metric / source
Employment	New: construction (2-4 jobs/village), O&M (1 technician), paid verifiers	Retrofit crew + O&M; paid local verifiers at each milestone	Repurposing crew (diesel-to-hybrid); retained O&M roles	FTE per village; ILO job-year coefficients
Household welfare	Full kerosene/diesel	Partial saving from reduced	Reduced PLN diesel subsidy	WorldPop HH count × unit

	saving (~10-20% HH income); productive-use income uplift (cold storage, fish-drying); extended study hours	outages; reliability premium on productive equipment uptime	transfer to village tariff; lower household energy cost	saving; productive-use revenue survey
Environmental	Forest protected: 24 villages screened to policy pathway = area not encroached; avoided CO ₂ vs diesel counterfactual per prescribed RE system	Avoided CO ₂ from replacing partial diesel hours; deforestation pressure reduced where biomass feedstock is agricultural residue	Largest avoided-emission potential: full diesel displacement; PLTD-to-PLTG asset conversion reduces imported fuel dependency	tCO ₂ /yr × carbon price; WDPA + Hansen forest area; Layer 2 capex × emission factor
Electrification quality / reliability	Baseline = 0; target: Service Quality Index ≥ 0.8 (≥80% months lit above threshold) sustained over 12 months post-commissioning	Target: SQI uplift from observed baseline to ≥0.8; monsoon-season drop eliminated	Target: SQI ≥ 0.9 (hybrid system provides higher base reliability than diesel-only)	VIIRS monthly radiance time series; lit_fraction and rad_cv from Layer 1 pipeline

Table 1. Terang SCBA and environmental impact matrix. SQI = Service Quality Index (fraction of observation months where village radiance exceeds detection threshold). Values are indicative pending population join (WorldPop) and ground-truthing. PLTD = diesel power plant; PLTG = gas-fired equivalent; CO₂ basis: direct combustion EF 0.7 kgCO₂/kWh for diesel reciprocating engines (Clarke, Wei et al., 2022). Note: Annex III reports median lifecycle GHG ~820 gCO₂eq/kWh; we use the direct combustion factor). Employment multipliers: IRENA & ILO, Renewable Energy and Jobs: Annual Review, 2024. ILLUSTRATIVE fields (Figure 10 verification cost comparison): no published field-audit benchmark currently available for Maluku; figures are indicative pending formal quote from an M&E firm operating in eastern Indonesia.



ILLUSTRATIVE — underlying benchmarks not yet validated with sector partners

Figure 10. Unit economics of verification-as-a-service: cost of conventional field verification per village per milestone against the Terang satellite-verified equivalent, with the resulting margin and the ring-fenced share directed to the village reliability fund.

Two product lines. The dark villages are the accessibility case: new capacity, clear additionality, strong concessional appeal. The villages with unstable service are the commercial case: retrofit of failing systems, lower cost per unit, faster deployment, larger pipeline. The first proves the mechanism on the hardest cases; the second is where volume and return reside.

Business model. Terang is asset-light, with three revenue lines: origination fees for a qualified pipeline; verification-as-a-service charged per milestone, priced below the field audit it replaces; and data subscriptions for planners and utilities. Because the data is free and the analysis automated, marginal cost per additional village approaches zero, while the accumulating archive of ground-truth confirmations becomes an asset competitors cannot easily replicate. Verification also identifies villages where energy is used productively – for cold storage, ice-making, fish-drying, coconut processing or milling – rather than for lighting alone. Because productive use raises household earnings and therefore willingness and ability to pay, Terang treats it as a leading indicator of persistence and preferentially routes concessional capital to developers who bundle productive equipment with generation.

Social enterprise structure. To prevent drift toward serving financiers at the expense of communities, a fixed share of verification revenue is ring-fenced into a village reliability fund that pre-finances the repairs monitoring detects are needed. The commercial engine funds the social outcome rather than competing with it.

Sustainability. The model does not depend on grant funding: inputs are free, costs scale sub-linearly with coverage, and customers hold standing budgets for these functions. By screening sites against protected areas and intact forest before recommending them, it also avoids the conflict between energy expansion and conservation that has constrained renewable deployment across Indonesia.

Timeline

Phase	Description	Duration
Pilot Validation	Complete detection and resource screening for Maluku; add population, official electrification records and conservation layers	3 months
Ground-Truth & Community Pilot	Field-confirm flagged villages; recruit and train paid local verifiers; establish village energy committees	4 months
First Close – Aru Islands	Fund the 16-village demonstration tranche; select developers; agree the milestone schedule with financiers	6 months
Provincial Scale-Up	Extend to the full 64-village Maluku portfolio and open the retrofit line for villages with unstable service	12 months
National Replication	Deploy to Papua and Nusa Tenggara, where the recorded access gap is	18 months

	largest; integrate with national programmes	
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Regional Expansion	Adapt to the Philippines and other ASEAN archipelagic states; license verification to regional green finance facilities	24 months+
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Conclusion

Terang is a system that uses satellite images to find villages that don't have reliable electricity – even when official records say they do – and then helps get them real, lasting power by only releasing funding when the lights stay on, not just when they're switched on once. This one idea is what makes it succeed across all six areas.

On accessibility, Terang finds villages that official records miss – 111 in Maluku alone, found only after we fixed a method that was accidentally leaving out the smallest, most remote villages. On energy security, it pushes villages toward local power sources instead of shipped-in diesel, and it can spot when a village loses power – even from a storm – within weeks instead of months or years. Its business model is new because it turns monitoring into a paid service that costs less than what financiers already spend on manual checks. This same idea makes it community-driven: villagers are paid to confirm what satellites see, and their confirmation is actually needed for the system to work – not just a formality. It counts as real social entrepreneurship because the money-making part of the business directly funds the part that protects communities: a share of every payment goes into a fund that pays for repairs the monitoring finds are needed. And its financing is creative because it makes it affordable, for the first time, to keep checking that a system still works months and years later – not just once when it's built.

All of this comes from one simple decision: only pay when the lights stay on, and prove it with data no one can fake. Terang isn't a new fund, a new subsidy, or a new piece of technology. It's the missing link that lets the money, the technology, and the communities that already exist finally connect – and stay connected.

At its core, Terang is an integrated system that uses satellite images to identify villages without reliable electricity—even when official records

claim they are fully electrified—and ensures they receive real, lasting power by gating funding on lights staying on, not just being switched on once. This single principle of paying for sustained service verified by unalterable satellite data drives impact across six essential areas, bridging critical energy access and security gaps across Southeast Asia.

On accessibility, Terang looks beyond main lands and macro-initiatives like the ASEAN Power Grid to serve the archipelagic communities beyond the reach of standard power lines. Using an unmasked polygon reduction method, it recovers remote island settlements lost in standard spatial models, pinpointing 111 dark-but-settled villages across Maluku alone. On energy security, it transitions vulnerable islands away from price-volatile, shipped diesel toward resilient local renewables or hybrid energy systems. Continuous orbital observation acts as a near-real-time early warning system, detecting storm-induced power outages within weeks rather than months or years.

Terang introduces a novel, asset-light business model that turns continuous oversight into a paid service, slashing verification overhead by up to 70% compared to manual field audits. This system makes monitoring genuinely community-driven by hiring local villagers as paid verifiers whose ground-truth confirmations are a required trigger for financial releases. Operating as a true social enterprise, a ring-fenced share of commercial verification revenue is automatically funneled into a reliability fund to pre-finance hardware repairs flagged by satellite tracking.

Finally, Terang unlocks creative financing by aggregating individual, unbankable mini-grids into district portfolios sized for institutional blended finance. Under Layer 3, capital structures are paired with Social Cost Benefit Analysis (SCBA) and Environmental Benefit Analyses, quantifying household fuel savings, local job creation, and carbon offsets, to deliver a fully de-risked pipeline for impact investors. Replicable across ASEAN island nations like the Philippines using open, global datasets, Terang is neither a new capital fund, a government subsidy, nor a hardware product; it is the missing verification link that allows capital, technology, and communities to finally connect and stay connected.

Contact

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Appendices

Mock Up

The demonstration application presents the pipeline in three linked views, built on real Maluku data. View 1, Gap Map: a province-wide map with every village coloured by class – dark but settled, intermittent, stable or unresolved – where selecting a village shows the official claim against the observed light record and a 22-month history. View 2, Prescriptions: the same map recoloured by recommended technology, each card stating technology, indicative capacity, cost band and the decision rule that produced it, so no output is a black box. View 3, Portfolio and Milestones: district bundles with aggregate cost and technology mix, a milestone board per village, and the verification chart showing monthly radiance against the detection threshold with the detected commissioning step and disbursement release.

[FIGURE A1 – INSERT IMAGE SET (4)]

Caption: Demonstration application, four screenshots: (i) View 1 Gap Map at province zoom; (ii) View 1 with a single dark village selected, showing its 22-month radiance history; (iii) View 2 Prescriptions with a recommendation card and its decision rule visible; (iv) View 3 Portfolio and Milestones showing a district bundle and the verification chart.

Source: Terang Streamlit demonstration application, running on real Maluku data.

Note: *Confirm the amber SYNTHETIC DATA badge is absent before capturing. Capture at 2x device pixel ratio.*

Working Flowchart

The pipeline runs in two stages. In Stage A, village boundaries, satellite radiance and settlement data are combined in a cloud geospatial platform: statistics are computed per village, the threshold calibrated from uninhabited reference points, villages classified, and flagged villages passed through resource and conservation screening. In Stage B, the application reads those exports and presents the three views. Milestones are evaluated by the same functions used in analysis, so the logic shown to users is identical to the logic used to classify. Because analysis runs once against the archives and the application reads only exported results, it stays fast and works without network access.

Flow: village boundaries and satellite imagery → radiance statistics and threshold calibration → classification → resource and conservation screening → technology prescription → portfolio bundling → milestone verification → disbursement.

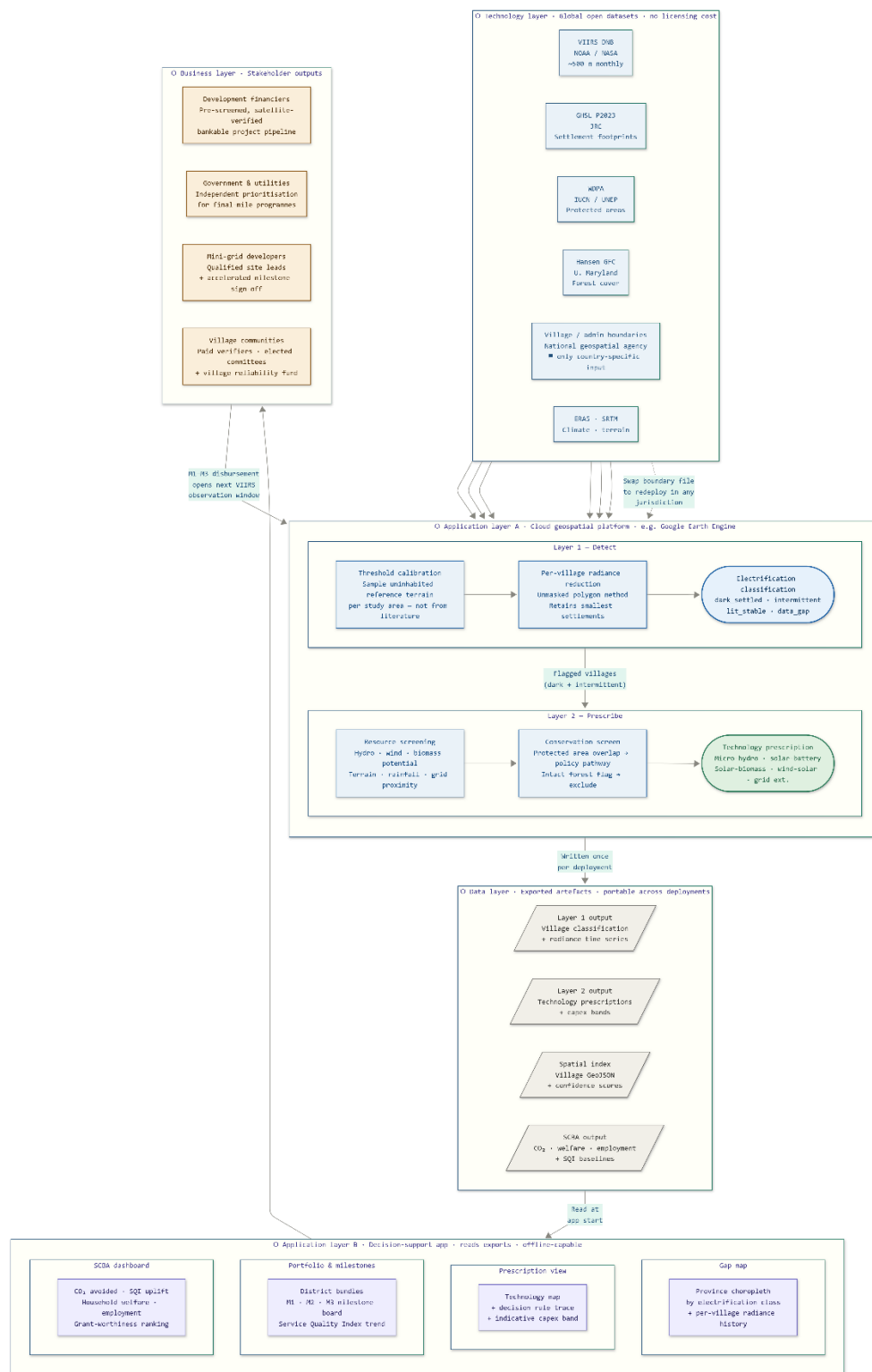


FIGURE A2. End-to-end pipeline flowchart. Stage A in the cloud geospatial platform: village boundaries and satellite imagery → radiance statistics and threshold calibration → classification → resource and conservation screening →

technology prescription. Stage B in the application: portfolio bundling → milestone verification → disbursement.

Financial Feasibility

The 73-village off-grid portfolio requires \$14.6 million, structured to match risk to return: a 40% non-repayable grant absorbs first-loss exposure, 25% guarantee-backed mezzanine capital targets 7% annually, and 35% senior capital – the tranche available to commercial investors – targets 13% annually. Capital is called as villages are actually built, not upfront, so investor money is never sitting idle ahead of construction.

Returns are grounded in real mini-grid benchmarks (ESMAP), not assumed: village-level revenue reflects actual generation capacity, a realistic tariff, and a collection rate that accounts for non-payment – the risk most rural electrification models ignore. We tested three outcomes, not one:

- Base case: senior investors earn 15.1%, mezzanine 7.1% – both above target.
- Upside case: senior 15.9%, mezzanine 8.2%, with comfortable debt-service coverage (2.89x).
- Downside case: the structure is built to fail safely. If costs run against the portfolio, capital exposure is capped – investors are not asked to keep funding a failing asset indefinitely. Losses stop and a distressed recovery is taken instead.

One caution we want investors to see clearly, not discover later: in the base case, debt-service coverage dips to 0.82x in the weakest year before recovering – the ten-year return is sound, but the path there is not perfectly smooth.

The case for this investment is not that it is risk-free – it is that the risk is bounded, priced, and disclosed. At base case, senior capital clears a 15% return; even in the scenario where operations underperform, the structure is designed to limit downside rather than compound it. We are asking investors to fund a portfolio where the upside is real and the downside has a floor – and where every assumption behind both is stated plainly enough to be checked.

Dark villages screened, Maluku pilot	111
Micro-hydro	32
Solar-biomass hybrid	31
Solar-battery	8
Wind-solar hybrid	2
→ Tractable off-grid villages financed by this portfolio	73
Grid extension (PLN mandate – not financed by this portfolio)	14
Policy pathway (protected area – no near-term financing)	24
→ Consistency check (should equal total dark villages screened above)	111

B. VILLAGE-LEVEL PHYSICAL ECONOMICS

Base mini-grid capex per village (\$)	\$200.000
Village mini-grid capacity (kW firm)	55,0
→ Implied cost per kW firm, this deal (\$) – check vs. ESMAP ~\$3,660/kW	\$3.636
Load factor, household-served mini-grid (%)	22,0%
→ Annual energy generated per village (kWh)	105.996

C. SCENARIOS – downside / base / upside, for the variables that actually drive risk

	Downside	Base	Upside
Tariff (\$/kWh)	\$0,22	\$0,30	\$0,40
Collection rate (%)	70,0%	90,0%	95,0%
O&M cost (% of capex per year)	12,0%	8,0%	6,0%

Village derating – share of villages underperforming/failing (%)

Capex overrun vs. base estimate (%)

Terminal capital recovery at exit (% of capex)

20,0%	10,0%	5,0%
20,0%	0,0%	-5,0%
70,0%	90,0%	100,0%
\$1.500,00	\$4.000,00	\$6.000,00

DEFAULT / WALK-AWAY MECHANICS – no real investor bails out a failing project forever

Default threshold – cumulative injection as % of senior capital

50,0%

Distressed terminal recovery if defaulted (% of capex)

20,0%

D. CAPITAL STRUCTURE

Output-based grant share (% , non-repayable – no return expected)

40,0%

Guarantee-backed mezzanine share (%)

25,0%

Commercial senior share (%)

35,0%

Senior target coupon rate (%/year)

13,0%

Mezzanine target coupon rate (%/year)

7,0%

SCENARIO COMPARISON SUMMARY

	Downside	Base	Upside
Net village cash flow per year, after derating (\$)	(\$4.941)	\$14.957	\$32.565
Senior IRR	-22,5%	15,1%	15,9%
Mezzanine IRR	-17,8%	7,1%	8,2%
Senior NPV, discounted at target rate (\$)	(\$6.373.969)	\$379.670	\$535.068

Mezzanine NPV, discounted at target rate
(\$)

Minimum DSCR across 10 years

(\$2.865.869)	\$11.054	\$226.587
-4,67x	0,82x	2,89x

Team Composition



Kartiko Cokro Sewoyo — technical lead. GIS Analysis Team Leader on the Budget Execution Data Analytics task force at Indonesia's Ministry of Finance, where he built a satellite change-detection system verifying whether government-funded agricultural programmes produced real outcomes on the ground.

MSc in Data Science, University of Manchester. Responsible for the detection pipeline, resource screening, milestone verification logic and the demonstration application.



Meiliza Fitri — energy and financing lead. State Asset Analyst at the Directorate General of State Assets Management, Ministry of Finance, with experience in energy asset valuation, policy research, and state asset optimization, including coal mining and oil and gas infrastructure assets. M.Eng. in Energy and

Resources Engineering from Chonnam National University, South Korea, and former Lecturer in Mining Engineering at Jambi University. Author and presenter of multiple international publications on energy transition, CBM reservoirs, EOR, LNG infrastructure repurposing, and state asset policy. Responsible for energy technology benchmarks, integration of energy models, financing structures, and regulatory policy framing.



Dwiky Anggawan Pradiptya Putra — policy and institutional lead. Policy Analyst at the Directorate General of State Assets Management, Ministry of Finance, with prior work on state asset revaluation, the simplification of five ministerial regulations into a single framework (PMK 173/PMK.06/2020), and analysis

of General Allocation Fund spending patterns for the Directorate General of Fiscal Balance. Master of Public Policy, University of Michigan (Ford School), and a UNICC-Columbia Thinkathon 2024 finalist. Responsible for policy pathway design, regulatory alignment with PLN and ESDM frameworks, and the institutional adoption route into Indonesian public finance.

The team combines geospatial and data science capability, energy sector and asset optimisation expertise, and direct policy access within Indonesian public finance – with a working relationship across three directorates general of the Ministry of Finance. This provides both a route to primary data and a credible pathway to institutional adoption. Team roles and responsibilities have been divided to allow parallel work streams; all three members are committed to attending the Manila finals if selected.

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